Samuel M. D. Seaver, PhD

Computational Biologist: Finding signals in noisy systems — connecting data and tools in networks of every kind: molecular, metabolic, regulatory, social

Chicago, IL area · samseaver@gmail.com · 773-655-3316 ORCID 0000-0002-7674-5194 · Google Scholar · GitHub · LinkedIn

---

Professional summary

Computational biologist who connects across systems — integrating data and tools at every scale to turn complex biology into solvable problems. 20+ years spanning molecular biology, bioinformatics, and quantitative modeling, and a decade building the open-source metabolic-modeling resources the plant- and microbial-systems-biology community relies on: first or senior author of the **ModelSEED Biochemistry Database** and **PlantSEED**, and a core member of DOE's **KBase** plants team building the apps and data integration for plant metabolic modeling. Work cited **~2,900 times** (h-index 13). Most recently adopting **machine learning** as a tool to make these models predictive from multi-omics data — physics-informed, mass-balance-constrained optimization (neural-network solver machinery, not trained black boxes). Seeking a staff-scientist role with the research independence to build the next generation of predictive, mission-driven metabolic modeling.

Research interests

Genome-scale metabolic modeling · constraint-based reconstruction & analysis · biochemistry database curation & thermodynamics · physics-informed / mechanistic machine learning · multi-omics integration · plant & microbial systems biology · bioenergy & sustainable-agriculture applications.

Education

- **PhD, Biological Sciences**, Northwestern University, 2011. Dissertation: *Modeling of gene regulation and metabolism in microbial organisms* (Amaral complex-systems group).
- **Postdoctoral Fellow**, Argonne National Laboratory, 2011–2015. Plant genomics & metabolic modeling.
- **MS, Molecular Modeling & Bioinformatics**, Birkbeck College, University of London, UK, 2001.
- **BS, Molecular Biology & Biochemistry**, Durham University, UK, 2000.

Appointments

- **Computational Scientist**, Argonne National Laboratory, Data Science & Learning Division, 2019–present.
- **Senior Scientist at-large**, University of Chicago, 2025–present.
- **Assistant Computational Scientist**, Argonne National Laboratory, 2015–2019.
- Scientist at-large, University of Chicago, Consortium for Advanced Science & Engineering, 2013–2025.
- Earlier: Intern, Integrated Genomics (2008); Research Technician (2002–2004) & Junior Bioinformatician (2001–2002), Northwestern University; Intern, European Bioinformatics Institute (EBI), Hinxton, UK (2000).

Selected accomplishments (impact highlights)

- **Created and maintain reference community resources.** Led development of the ModelSEED Biochemistry Database (*NAR* 2021) and PlantSEED (*PNAS* 2014; *Plant Journal* 2018) — the resources thousands of labs use to reconstruct and compare genome-scale metabolic models — and, as a core member of the KBase plants team, built the plant metabolic-modeling apps and data integration in DOE's KBase (*Nature Biotechnology* 2018).
- **Brought machine learning to metabolic modeling.** Senior-authored a flexible neural-mechanistic hybrid that simulates the iron-deficiency response in plant plastidial metabolism (bioRxiv 2025) — physics-informed neural-network solver machinery coupled to a mechanistic model (no trained weights).
- **Delivered high-throughput reconstruction at scale.** Co-developed ModelSEED v2 for high-throughput genome-scale model reconstruction across phylogenetically diverse organisms.
- **Curated biochemistry at scale.** Led structure and formula curation across ~9,000 reactions / ~6,500 compounds underpinning ModelSEEDTemplates v7.0 and PlantSEED, including a mass-balance exclusion mechanism for R-group compounds and stereochemistry-preservation safeguards.
- **Integrated demanding multi-omics datasets.** Built and verified pipelines integrating transcriptomics, proteomics, and dual-provider metabolomics across a four-species × two-tissue circadian time-course, mapped onto genome-scale metabolic networks.
- **Built institutional AI-development infrastructure.** Created the onboarding path for AI-assisted scientific computing through Argonne's internal LLM gateway (on-network, credential-compliant), including HPC/Jupyter tool integrations now used by lab staff.

Grants & funding

Amounts are total anticipated project value; role noted per award. Source: DOE Current & Pending Support (2025).

As Principal Investigator

- **Integration of computational tools to explore the diversity of temporal regulation in plant specialized metabolism.** DOE Office of Science (BER, Genomic Science Program), PRJ1011480. **PI.** 2023–2026. **$1,041,228.** Collaborative with K. Greenham (Univ. of Minnesota); develops ML + 'omics + metabolic reconstruction to predict metabolite abundances from enzyme abundances.
- **Quantitative Plant Science Initiative (QPSI) — Argonne computational subaward.** DOE BER Science Focus Area led by C. Blaby-Haas, Brookhaven National Laboratory. **PI of Argonne subaward, $450,000, 2022–2023.** Led the metabolic-reconstruction and simulation aim on Fe/Zn metal-stress response in poplar and sorghum, within the multi-institution QPSI SFA.

As Co-Investigator / Senior Personnel (funded)

- **KBase: An Integrated Knowledgebase for Predictive Biology and Environmental Research.** DOE, PRJ1003559. Senior personnel. 2021–2026. $16,420,000 (total program).
- **Unlocking the Molecular Basis of Plant-Pathogen Interactions to Create Resilient Bioenergy Crops.** DOE, PRJ1011481. Senior personnel. 2023–2026. $1,080,000.

Pending

- A Hybrid Machine Learning–Mechanistic Prediction Framework for Harnessing Microbial Synergy to Enhance Switchgrass Growth for Biofuel Production. DOE BER (San Francisco State Univ.). 2025–2030. $1,655,975.
- Integrative modeling for predicting mediators that drive variation in diel sulfur flux during heat stress in pennycress accessions. DOE BER (Univ. of Minnesota). 2025–2030. $946,348.
- Enabling Sorghum Gene Functional Characterization for Efficient Use of Water. DOE (Purdue Univ.). 2025–2028. $145,310.

Grant-writing record: PI on 2 funded DOE awards; senior/key personnel on multiple funded programs (incl. the $16.4M KBase program); 3+ additional proposals submitted to DOE-SC/BER as PI or co-investigator.

Publications

Metrics: **2,914 citations · h-index 13 · i10-index 16** (Google Scholar, Aug 2026). † first author · ‡ senior/corresponding author.

Reference resources & infrastructure

1. **Seaver SMD**†, Liu F, Zhang Q, Jeffryes J, Faria JP, Edirisinghe JN, Mundy M, Chia N, Noor E, Beber ME, Best AA, DeJongh M, Kimbrel JA, D'haeseleer P, McCorkle SR, Bolton JR, Pearson E, Canon S, Wood-Charlson EM, Cottingham RW, Arkin AP, Henry CS. The ModelSEED Biochemistry Database for the integration of metabolic annotations and the reconstruction, comparison and analysis of metabolic models for plants, fungi and microbes. *Nucleic Acids Research* 49(D1):D575–D588 (2021). *336 citations.*
2. Arkin AP, Cottingham RW, Henry CS, Harris NL, Stevens RL, Maslov S, … **Seaver SMD**, … Yu D. KBase: The United States Department of Energy Systems Biology Knowledgebase. *Nature Biotechnology* 36(7):566–569 (2018). *1,917 citations.* (KBase consortium, ~70 authors.)
3. **Seaver SM**†, Gerdes S, Frelin O, Lerma-Ortiz C, Bradbury LM, Zallot R, Hasnain G, Niehaus TD, El Yacoubi B, Pasternak S, Olson R, Pusch G, Overbeek R, Stevens R, de Crécy-Lagard V, Ware D, Hanson AD, Henry CS. High-throughput comparison, functional annotation, and metabolic modeling of plant genomes using the PlantSEED resource. *PNAS* 111(26):9645–9650 (2014). *101 citations.*
4. **Seaver SMD**†, Lerma-Ortiz C, Conrad N, Mikaili A, Sreedasyam A, Hanson AD, Henry CS. PlantSEED enables automated annotation and reconstruction of plant primary metabolism with improved compartmentalization and comparative consistency. *The Plant Journal* 95(6) (2018). *49 citations.*
5. **Seaver SM**†, Bradbury LM, Frelin O, Zarecki R, Ruppin E, Hanson AD, Henry CS. Improved evidence-based genome-scale metabolic models for maize leaf, embryo, and endosperm. *Frontiers in Plant Science* 6 (2015). *75 citations.*
6. Wood-Charlson EM, Henry CS, Dehal PS, Mahmud G, Allen BH, Beilsmith K, … **Seaver S**, … Arkin AP. KBase: Open-source platform for collaborative biological data analysis and publication. *Journal of Molecular Biology* 438(18):169676 (2026). (KBase consortium, ~42 authors.)

Machine learning & high-throughput modeling

7. El Alaoui S, Henry CS, Blaby-Haas CE, Paape T, Xie M, **Seaver SM**‡. Simulating Iron Deficiency in Plant Plastidial Metabolism With a Flexible Neural-Mechanistic Hybrid Approach. *bioRxiv* 2025.06.10.658179 (2025).
8. Faria JP, Liu F, Edirisinghe JN, Gupta N, **Seaver SMD**, Freiburger AP, et al. ModelSEED v2: High-throughput genome-scale metabolic model reconstruction with enhanced energy biosynthesis pathway prediction. *bioRxiv* (2023). *35 citations.*
9. Faria JP, Liu F, Edirisinghe JN, Gupta N, **Seaver SMD**, Freiburger A, et al. Advancements in Genome-Scale Model Reconstruction facilitate comparative analysis of microbial energy biosynthesis, auxotrophy, and essential cofactors. *9th Conf. on Constraint-Based Reconstruction and Analysis* (2024).

Applied metabolic modeling & multi-omics

10. Mishra A, Bhat A, Kumari S, Sharma R, Braynen J, Tadesse D, El Alaoui S, **Seaver S**, Grosjean N, Ware D, Xie M, Paape T. Time-series multi-omics analysis of micronutrient stress in *Sorghum bicolor* reveals iron and zinc crosstalk and regulatory network conservation. *Plant Biology (Stuttg)* (2025). PMID 40402192. *4 citations.*
11. Beilsmith K, Henry CS, **Seaver SMD**‡. Genome-scale modeling of the primary–specialized metabolism interface. *Current Opinion in Plant Biology* 68 (2022). *12 citations.*
12. **Seaver SMD**†. Systems-level analysis of the plasticity of the maize metabolic network reveals novel hypotheses in the nitrogen-use efficiency of maize roots. *Journal of Experimental Botany* 73(1) (2022). *2 citations.*
13. Kumari S, Kumar V, Beilsmith K, **Seaver SMD**, Canon S, Dehal P, Gu T, Joachimiak M, Lerma-Ortiz C, Liu F, Lu Z, Pearson E, Ranjan P, Riel W, Henry C, Arkin A, Ware D. A KBase case study on genome-wide transcriptomics and plant primary metabolism in response to drought stress in Sorghum. *Current Plant Biology* 28:100229 (2021). *11 citations.*
14. Jeffryes JG, **Seaver SMD**, Faria JP, Henry CS. A pathway for every product? Tools to discover and design plant metabolism. *Plant Science* 273 (2018). *27 citations.*
15. Noutsos C, Perera AM, Nikolau BJ, **Seaver SM**, Ware DH. Metabolomic profiling of the nectars of *Aquilegia pubescens* and *A. canadensis*. *PLoS One* 10(5) (2015). *26 citations.*

16. Töpfer N, **Seaver SMD**, Aharoni A. Integration of plant metabolomics data with metabolic networks: progresses and challenges. *Plant Metabolomics: Methods and Protocols (Methods Mol Biol)* (2018). *17 citations.*
17. Faria JP, Khazaei T, Edirisinghe JN, Weisenhorn P, **Seaver SMD**, et al. Constructing and analyzing metabolic flux models of microbial communities. *Hydrocarbon and Lipid Microbiology Protocols* (2016). *18 citations.*
18. Colasanti R, Edirisinghe JN, Khazaei T, Faria JP, **Seaver S**, Xia F, et al. Tapping the Wealth of Microbial Data in High-Throughput Metabolic Model Reconstruction. *Metabolic Flux Analysis: Methods and Protocols* (2014).

Foundational reviews & early quantitative work

19. Gerdes S, Lerma-Ortiz C, Frelin O, **Seaver SM**, Henry CS, de Crécy-Lagard V, Hanson AD. Plant B vitamin pathways and their compartmentation: a guide for the perplexed. *Journal of Experimental Botany* 63(15) (2012). *122 citations.*
20. **Seaver SM**†, Henry CS, Hanson AD. Frontiers in metabolic reconstruction and modeling of plant genomes. *Journal of Experimental Botany* 63(6):2247–2258 (2012). *92 citations.*
21. **Seaver SM**†, Sales-Pardo M, Guimerà R, Amaral LA. Phenomenological model for predicting the catabolic potential of an arbitrary nutrient. *PLoS Computational Biology* 8(11):e1002762 (2012). *PMCID PMC3486842.*
22. **Seaver SM**†, Moreira AA, Sales-Pardo M, Malmgren RD, Diermeier D, Amaral LA. Micro-bias and macro-performance. *European Physical Journal B* 67(3):369–375 (2009). *10 citations.*
23. Salerno WJ, **Seaver SM**, Armstrong BR, Radhakrishnan I. MONSTER: inferring non-covalent interactions in macromolecular structures from atomic coordinate data. *Nucleic Acids Research* 32(Web Server):W566–W568 (2004). *47 citations.*
24. **Seaver SMD**†. Modeling of gene regulation and metabolism in microbial organisms. PhD dissertation, Northwestern University (2011).

Selected conference abstracts (PlantSEED in KBase; Integrated Transcriptomics with KBase, PAG XXVI/XXVII; AIChE 2017; social- and collaboration-network work, 2006) available on request.

Software & scientific resources

- **ModelSEED Biochemistry Database** — reference biochemistry DB (compounds, reactions, structures, thermodynamics, mass/charge balancing); ongoing curation lead.
- **PlantSEED** — automated plant-metabolism annotation & genome-scale reconstruction; integrated in KBase; leading a v3 refactor.
- **KBase (plants team)** — built plant metabolic-modeling apps and integrated plant data on DOE's Systems Biology Knowledgebase.
- **BioFlux — neural-mechanistic hybrid modeling** — biochemistry-informed mechanistic gradient optimization with mass-balance constraints (PINN solver machinery; no trained weights).
- **Argonne AI-development infrastructure** — internal-LLM-gateway onboarding, reverse proxy, and HPC/Jupyter MCP integrations for AI-assisted scientific computing.

Mentoring & leadership

Have trained **10+ interns, graduate students, and postdocs** over ~15 years, leading by growing people into new capabilities and building the shared tools and teams that let them do their best work. Recent examples: senior-authored the flagship ML preprint led by a mentee (S. El Alaoui); supervised graduate and intern researchers on the structure-validation and thermodynamics pipelines.

Selected collaborations

KBase consortium (DOE, multi-lab) · Greenham lab (circadian multi-omics) · DOE Joint Genome Institute (plant genomics/metabolomics) · Hanson lab (plant biochemistry) · long-standing collaboration with C. S. Henry

(ModelSEED/KBase).

Invited talks & seminars

- **Invited talk (2026)** — *Interpreting metabolic imbalance in hybrid ML–constraint-based models* — 10th Conference on Constraint-Based Reconstruction and Analysis (COBRA), Potsdam, Germany.
- **Invited talk (2025)** — *Science, Unmuted* (accessibility & inclusion in science) — University of Tennessee, Knoxville (REU program).
- **Invited departmental seminar (2024)** — *Predicting metabolic fluxes* — University of Minnesota, Dept. of Plant & Microbial Biology.
- Invited talk (with N. Töpfer) — **German Conference on Bioinformatics (GCB)**, online (2021).
- Invited talk — plant-bioinformatics conference at **IPK Gatersleben**, Germany (2019).
- Oral presentation — **ICAR 2018** (Int'l Conference on Arabidopsis Research), Turku, Finland — plant metabolic networks.
- **Invited European seminar series (2015)** — *"The key evidence underlying the top-down construction, propagation and analysis of genome-scale plant metabolic models"* — presented at the Max Planck Institute of Molecular Plant Physiology (Potsdam), CEPLAS / Cluster of Excellence on Plant Science (Düsseldorf), Institut Jean-Pierre Bourgin (Versailles), Vrije Universiteit Amsterdam, Oxford Brookes University, and Universitat Rovira i Virgili (Tarragona).
- *PlantSEED Enables Automated Reconstruction of Plant Primary Metabolism…* — Charles University, Prague (2018).
- *The key evidence underlying the construction and analysis of plant-microbial metabolic models* — Plant Systems Biology Group, Oak Ridge National Laboratory (2016).
- *Plant Workflows in Persistent KBase Narratives* — Multi-scale Plant Modeling Workshop, EMSL, Pacific Northwest National Laboratory (2016).
- *Enabling the Systemic Inter-specific Comparison of Predicted Metabolic Responses…* — Plant Biology 2016, Austin, TX.
- *When Biologists and Modelers Meet in KBase: Modeling Plant–Microbe Metabolic Interactions* (with D. Weston) — Plant & Animal Genome (PAG) XXIV, San Diego (2016).
- *Metabolic Modeling of Plant–Microbe Interactions in KBase* and *The PlantSEED Resource for Functional Annotation and Cross-Kingdom Comparative Genomics* — PAG XXIII, San Diego (2015).
- *Refinement of Maize Genome-scale Metabolic Models for Leaf, Embryo, and Endosperm* — Plant Biology 2015, Minneapolis, MN.
- *Comparative Genomics and Systems Approaches to Primary Metabolism in Magnoliophyta* — Plant Biology 2013, Providence, RI; and related talks at PAG XXII (2014) and the Symposium on Plant Science & Agricultural Biotechnology, NC State (2013).

Workshops & symposia organized

- **PlantSEED Workshops** — Northwestern University (2016, 2018) and University of Florida (2015, 2017).
- **Metabolic Networks Symposia/Workshops** — Plant Biology 2013 (Providence), Plant Biology 2016 (Austin), and ICAR 2018 (Turku, Finland).
- **KBase & metabolic-modeling workshop** — ASPB Plant Biology (online, 2021), co-led with K. Beilsmith.

Professional service & activities

- Community workshops: Nitrogen Economy "Big Idea" Workshop, ORNL (2016); iPASS Multiscale Plant Modeling Workshop, PNNL (2016); Software Carpentry instructor, Plant Biology 2014 (Portland).
- Member, American Society of Plant Biology (2011–2015).

Honors & awards

- NIH Predoctoral Biotechnology Training Grant (T32), 2006–2008.
- Student Poster Award, Northwestern Institute for Complex Systems, 2006.

Technical skills

Python · R · genome-scale/constraint-based modeling (COBRA, FBA) · physics-informed neural networks · multi-omics integration (edgeR/TMM, proteomics, metabolomics) · cheminformatics (InChI/SMILES) · comparative genomics (DIAMOND/BLAST orthology) · scientific database engineering · reproducible HPC pipelines · large open-source project leadership.